KANERVA: Token-Budgeted, Validity-Aware Memory for Coding Agents, Without an LLM in the Loop

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Abstract

Coding agents forget everything between sessions, and the memory systems built for them mostly select context by similarity top-k, ignore what items cost in tokens, and have no model of whether a stored fact is still true. We cast agent memory as online budgeted selection, show that the surveyed systems are special cases, and derive where each is worst-case suboptimal. We then present KANERVA, a local memory layer with no LLM in the loop. It scores each candidate with a calibrated relevance model over retrieval, item and context features, selects under a hard token budget, keeps an item's *validity* apart from its *usefulness*, and expands candidates through neighbouring turns. We prove a $\frac{1}{2}(1 - \frac{1}{e})$ selection guarantee with a per-query certificate, a bound on how probability error reaches the selection, a dynamic regret bound that separates the cost of expiring items, a linear lower bound for validity-blind learners, a rate–distortion bound for reversible consolidation, and a stationary store-size result. On a held-out test with no LLM in the evaluation, averaged over budgets from 128 to 4096 tokens, KANERVA recovers 0.566 of the evidence on BEAM against 0.478 for hybrid retrieval with a cross-encoder and 0.440 for hybrid alone, with no cross-encoder of its own. On RepoMem, a new benchmark built from real repository histories, it reduces injection of outdated facts from 0.997 to 0.194 without structured keys, though a simple keep-the-newest-twin rule does as well, and neither helps on natural-language updates. There its recall cannot be told apart from hybrid retrieval at most budgets and is behind it at large ones, the lighter configurations that need no neural model do worse, online learning adds nothing on top of the fitted model, and forgetting costs recall.

1 Introduction

A coding agent starts each session knowing nothing it learned in the last one. Native memory is a set of files loaded whole: Claude Code loads its `CLAUDE.md` files and the first 200 lines or 25 KB of a memory index, with no ranking by the task at hand [4]. Memory products add retrieval, but they select by similarity top-k or leave the choice to the agent, and they spend LLM calls on every write. The published evidence does not show that this is good enough. On LoCoMo, Mem0's own table has a full-context baseline ahead of every memory system tested [12]. Retrieval method moves accuracy far more than write strategy, and raw chunks with no LLM calls match fact extraction [69]. On coding tasks, unfiltered experience gives little or negative benefit [75], and eleven of twelve pairings of solvers with memory systems fail to beat memory-off [16].

These findings point at the read path, and at three defects in it. Selection ignores what an item costs in tokens. It ignores what the other selected items already say. And it ignores whether an item is still true, which is the common failure in a codebase, where a function signature read last month may have changed since.

This paper treats memory as an online budgeted selection problem and makes four contributions.

1. A framework in which the surveyed systems are special cases, with elementary worst-case arguments for what each one gives up (Section 3).

2. A method that scores each candidate with a calibrated relevance model and selects under a hard token budget, with validity modelled apart from usefulness and estimated from the store's own version history (Section 4). It runs locally and needs no LLM.

3. A regret decomposition for items that expire (Theorem 2). Stationary linear submodular bandit results exist [57, 68, 70]; they exclude this case. Here non-stationarity costs the accumulated validity error, while a validity-blind learner has linear regret (Proposition 9). Proposition 7 bounds how probability error reaches the selection, and the other theory results are a rate–distortion bound for reversible consolidation and a stability result for store size.

4. RepoMem, a coding-memory benchmark built from the real histories of Python repositories, in

*Code and data pointers: `https://github.com/katolikov/kanerva`.

which facts change for real and every label comes from the code by static analysis. We evaluate on held-out RepoMem repositories and held-out BEAM conversations against a dozen reading baselines, with ablations, confidence intervals and the cases where the method loses.

Scope of the evidence. All results here are *retrieval-level*: evidence recall under a token budget, tokens injected, staleness errors, latency and store size. No LLM was available for these experiments, so we report no answer accuracy and no end-to-end task success. Recall under a budget is a necessary condition for those outcomes, not a measurement of them. A controlled study found that context strategy moved correctness on coding tasks by no more than 10 to 15 points [28], so large end-to-end gains should not be assumed.

2 Related work

Memory systems. Memory layers include OS-style paging by the agent [43], LLM-curated fact stores [12], temporal knowledge graphs with bi-temporal edge invalidation [49], linked notes [67], graph retrieval [15, 20, 21], and recursive summary trees [51]. Generative Agents scores memories by recency, importance and relevance [45]. Experience memories such as Reflexion, Voyager, ACE and ReasoningBank keep lessons or skills [42, 52, 63, 71]. All of these use a write path with LLM calls, a read path with a fixed k or an agent decision, or both. Of these systems, Zep/Graphiti is the one with a mechanism for validity. It is binary and depends on the extractor producing the invalidating edge.

Budgeted selection. Token-budgeted submodular selection for context is an active preprint line. PACMS applies facility location under a knapsack as a pluggable engine and lists learned relevance as future work [19]. AdaGReS scores relevance against redundancy [46], and PC-SubMax compresses prompts with a regularized objective [74]. The theory rests on Nemhauser et al. [41], on budgeted maximum coverage [29], on partial enumeration [17, 56], and on cost-benefit greedy with lazy evaluation [30, 34, 39]. MMR [8] and DPPs [31] target diversity without a budget in tokens.

Learning. Combinatorial and linear semi-bandits give regret with approximate oracles [9, 10, 33, 64], and linear submodular bandits extend this to diversified retrieval and to knapsacks [57, 68, 70]. Our analysis uses the confidence sets of OFUL [1]. In agent memory, DeMem proves regret bounds for a K-state memory partition [76] and MemRL shows Q-value convergence [72]; RSCB-MC and MERA use bandits to choose agent actions in coding [24, 25]. None of these treats token-budgeted selection of items whose validity changes. On reward signals, self-scored episodes inflate wrong answers [5], so we use execution signals.

Validity and forgetting. ACT-R base-level activation models the power law of forgetting [2, 3], and it has been applied to LLM memory [22]. Staleness is a measured failure: with a budget of two records, models acted on stale records in most episodes of one controlled study [40], and conflict benchmarks exist for chat [59]. Consolidation follows complementary learning systems [32, 38]. Rate–distortion views of compaction appear in [13, 53, 76]. Colaco and Lahjouji argue that repeated irreversible summarization degrades end-task error faster than retrieval-backed memory, which agrees with the context collapse reported for ACE [71], and motivates keeping sources.

Compression and parametric memory. Text compression such as LLMLingua [26, 27, 44] works with closed APIs and is complementary to selection. Gist tokens, AutoCompressors, ICAE, and KV-cache eviction [11, 18, 36, 47, 66, 73] and test-time weight updates [6, 7, 54] need access to model internals that closed APIs do not expose.

Benchmarks. LoCoMo [37], LongMemEval [65], MemoryAgentBench [23] and BEAM [60] cover conversational memory. SWE-ContextBench [75] and VibeMemBench [16] cover coding experience, MemoryCode covers multi-session coding instructions on synthetic data [48], and ContextBench covers retrieval within a task [35]. None of these, as far as we found, tests repository facts that change between sessions over real history.

3 A unified framework

Memory. At time t the store is a finite set M_t. An item is a tuple $m = (\text{text}, e, \tau^{\text{c}}, \tau^{\text{v}}, \pi, c, k, P)$: content, unit embedding, creation time, last time the item was confirmed, provenance, token cost $c(m) \in \mathbb{N}_{>0}$, an optional *slot key* k that identifies the fact the item states, and parent links P to the items it was consolidated from. Items sharing a key form a version chain.

Validity and usefulness. Fix a step context (query) q. Let $\rho_t(m) = \Pr[m \text{ still holds at } t]$ and $\mu_q(m) = \Pr[m \text{ helps with } q \mid m \text{ holds}]$. The *effective weight* of an item is $w_q(m) = \rho_t(m)\,\mu_q(m)$. Separating the two matters: ρ moves with time and changes in the world, μ with the query. Section 5 shows that a learner with no age or version information has linear regret.

Selection. Injecting $S \subseteq M_t$ has utility $U(S \mid q)$, and the context budget is a hard constraint:

$$\max_{S \subseteq M_t} \mathbb{E}\left[U(S \mid q_t)\right] \quad \text{s.t.} \quad c(S) := \sum_{m \in S} c(m) \le B. \quad (1)$$

We study the family

$$F(S) = (1-\gamma)\Big[\sum_{f \in \mathcal{F}_q} \omega_f \Big(1 - \prod_{m \in S} \big(1 - w(m)\, a_{mf}\big)\Big) + \eta \sum_{m \in S} w(m)\Big] + \gamma \sum_{c \in C} \frac{p_c}{\sum_{c'} p_{c'}} \max_{m \in S} s(c,m). \quad (2)$$

Here $\mathcal{F}_q$ is a set of facets of the query with weights $\omega_f \ge 0$, $\sum_f \omega_f = 1$, and $a_{mf} \in [0,1]$ is the chance that item m answers facet f given that it is valid and useful. The first term is the expected fraction of the need that gets covered when items succeed independently, the second the expected number of useful items injected, and the third a facility-location coverage of the candidate pool C with similarity $s \in [0,1]$ and fixed importance p_c, which removes near-duplicates.

Lemma 1. *For fixed $w \in [0,1]^M$, F is non-negative, monotone and submodular, with $F(\emptyset) = 0$. It is coordinate-wise non-decreasing in w and $|F(S;w) - F(S;w')| \le L \sum_{m \in S} |w(m) - w'(m)|$ with $L = (1-\gamma)(1+\eta)$.*

The online problem. A memory policy $\Pi = (\mathsf{W}, \mathsf{R}, \mathsf{C}, \mathsf{F})$ decides what to write, what to read, what to consolidate and what to forget. For a sequence of steps it earns

$$J(\Pi) = \mathbb{E} \sum_{t=1}^{T} \Big[U(S_t \mid q_t) - \kappa_{\mathrm{r}}\, c(S_t)\Big] - \kappa_{\mathrm{w}}\, \mathrm{WriteCost}(\Pi) - \kappa_{\mathrm{s}} \sum_t c(M_t), \quad (3)$$

subject to $c(S_t) \le B$ for all t. Reading costs tokens on every step, writing costs LLM calls or latency, and the store costs retrieval time and noise. Learning, in this paper, is updating Π and M_t; model weights never change.

3.1 Existing methods as special cases

Table 1 places the surveyed systems in (1)–(3). Nearly all read policies reduce to the modular case $\eta = 1, \gamma = 0$, $\mathcal{F}_q = \emptyset$ with a cardinality constraint and a method-specific score. The systems differ mostly in what they store, which is the W component.

Parametric methods (test-time training [54, 55], memory layers [7], Titans [6], gist tokens and KV-cache eviction [36, 47, 73]) keep memory in weights or attention state. They fall outside the framework by construction, and they require write access to weights or to the cache, which a closed API does not expose.

3.2 Where the special cases are suboptimal

The following statements are worst-case and elementary; the point is that each identifies a design defect that a better method must remove. Proofs are in Appendix A. Write OPT for the optimum of (1).

Proposition 1 (Cost blindness)**.** *Any rule that fills the budget in the order of a score independent of c (top-k fill, MMR, reciprocal rank fusion, reranking) has worst-case ratio at most $1/\lfloor B/c_{\min} \rfloor$ against* OPT*, even for additive U.*

Proposition 2 (Redundancy blindness)**.** *Any modular rule has worst-case ratio at most $1/k$ on coverage objectives with k facets when one facet has at least k near-duplicate items.*

Proposition 3 (Recency as proxy)**.** *A relevant item m followed by later items of total cost above B, each no larger than m, is never injected by recency-only selection, however relevant it is, whereas a rule that scores by relevance injects it whenever it outranks those items.*

Proposition 4 (Time-invariant scoring)**.** *Any selector whose score is invariant under permuting the timestamps of items with equal content features injects an outdated version ahead of the current one with probability at least $1/2$ on version pairs of equal text similarity to the query.*

Proposition 5 (Fixed policies)**.** *For any fixed linear scoring vector θ there is a $\theta^\star$ for which the fixed policy earns regret $\Omega(T)$, while the learner of Theorem 2 has regret $\tilde{O}(\sqrt{T})$ on the same instance.*

Table 1 shows which propositions apply: every k-limited row is cost-blind, every modular row is redundancy-blind, every row with no validity model or a recency proxy falls under Propositions 3–4, and every non-learning row under Proposition 5. The systems with invalidation (Zep/Graphiti) avoid Proposition 4 only for facts where the extractor produced the invalidating edge.

4 Method

KANERVA is the combination of the components below, implemented with SQLite, FTS5 and NumPy. No component calls an LLM. When a host agent is present it can supply structured writes through tools, which is how semantic extraction happens at no added API cost.

4.1 Write path

A write redacts secrets, deduplicates against live items (an identical write only refreshes the confirmation time

Table 1: Surveyed methods as instances of the framework. "Score" is the w used by the read policy; k denotes a cardinality limit.

Method	Write W	Score w	Constraint	Validity ρ	Learns
Top-k RAG, hybrid, rerank [14, 50]	chunks	BM25/cosine/RRF/CE	k	none	no
MMR [8]	chunks	greedy rule, λ trade-off	k	none	no
PACMS, AdaGReS [19, 46]	chunks	facility location; relevance−redundancy	$c(S) \le B$	none	no
MemGPT/Letta [43]	agent tool calls	LLM decides paging	block sizes	none	no
Mem0 [12]	LLM add/update/delete	similarity	k	LLM rewrite	no
Zep/Graphiti [49]	LLM graph extraction	hybrid + rerank	k	$\rho \in \{0,1\}$ by edge invalidation	no
A-MEM [67]	LLM notes and links	similarity + links	k	LLM note evolution	no
HippoRAG 2 [21]	OpenIE triples	personalized PageRank	k	none	no
RAPTOR [51]	recursive summaries	similarity on all levels	k or tokens	none	no
Generative Agents [45]	observations, reflections	recency+importance+relevance	k	recency proxy	no
ACE, ReasoningBank [42, 71]	LLM-curated lessons	similarity or whole playbook	k	none	via text
Claude Code files [4]	model-decided notes	constant (query-independent)	200 lines/25 KB	none	no
Full history	everything	constant	window size	none	no
LLMLingua family [26, 44]	—	—	compresses $c(m)$	none	no

τ^{v}), and stores the item with its FTS5 entry and embedding. A write with a key closes the version interval of the live item with that key: the old item keeps its text and gets τ^{end} and a pointer to its successor. Nothing is rewritten, so history stays queryable.

4.2 Read path

Algorithm 1 shows retrieval. Candidates are the union of the top-200 lexical and dense hits, plus the items written next to the 50 best fused hits in the same session. Neighbours matter because a conversation often splits one fact across adjacent turns, and on our development data they added about twice as much evidence per candidate as a larger pool.

Relevance. Each candidate gets a calibrated probability of being needed, from gradient-boosted trees over 29 features in five groups: retrieval signals (ranks, scores, query-term coverage), the item itself (length, content-word density, question form, numbers), role and speaker (whether a first-person query meets a user turn, whether a named speaker appears in the query), query–item overlap (entities, numbers, and a question type such as "when" or "how many" against dates and numbers in the item), and context (similarity of neighbouring turns and of the session, recency, ACT-R base-level activation [3]). Three more features come from an optional local reranker. The trees are fitted on development data with equal weight per dataset and per query, exported to a compact array form and evaluated with NumPy in tens of milliseconds for a few hundred candidates, so the package has no learning library at run time. Separate models cover the lexical, static-embedding, dense and reranker configurations.

Objective. With calibrated weights, $F(S) = \sum_{m \in S} w_m$ is the expected number of needed items in S. This is the modular case of the objective of Section 3, so Theorem 1 still applies, and Proposition 7 says how probability error reaches the selection. With calibrated weights the coverage facets and the facility-location term had no measurable effect on development data, so they are off by default. Length enters twice, as a feature and as the cost, which is intended: the model can learn that short turns are rarely needed, so density no longer favours short junk once the strong candidates run out. This removes the main weakness of the first version of the method, a linear prior over retrieval signals alone, which lost to plain ranking at large budgets.

Only the density variant $\varrho = 1$ and the singleton carry the guarantee; the other runs can only raise F, because the maximum is taken over all runs.

The plugins add one step that the benchmarks do not exercise: they drop selected items whose probability is below 0.04, so that a query with nothing relevant injects little.

Algorithm 1 RETRIEVE(q, B, t)

$C \leftarrow$ top-N of FTS5 $\cup$ top-N of dense search $\cup$ same-session neighbours of the best fused hits
Remove superseded items, and items with a newer near-twin, unless q asks for history
for $m \in C$ **do**
  $p_m \leftarrow$ relevance model on features of (m, q); $x_m \leftarrow (p_m, u_m)$ ▷ u_m: past use
  $\mu_m \leftarrow \mathrm{clip}_{[0,1]}\big(x_m^\top \hat{\theta} + \beta \|x_m\|_{V^{-1}}\big)$
  $\rho_m \leftarrow$ Lomax survival (4) at $t - \tau^{\mathrm{v}}(m)$
  $\rho_m \leftarrow \rho_m \prod_{m' \text{ newer}} \big(1 - \pi\, g(s(m, m'))\big)$ ▷ no key
  $w_m \leftarrow \rho_m \mu_m$
end for
$\mathcal{S} \leftarrow \{\textsc{Greedy}(F, B, \varrho) : \varrho \in \{1, \frac{1}{2}\}\} \cup \{\arg\max_{c(m) \le B} F(\{m\})\}$
return $\arg\max_{S \in \mathcal{S}} F(S)$, and on request certificate (5)

4.3 Validity

For a class k of facts (for example, function signatures), the unknown change rate has a Gamma(α, β) prior. Af-

Algorithm 2 GREEDY(F, B, ϱ)

$S \leftarrow \emptyset$; $A \leftarrow \{m : c(m) \le B\}$
while $A \neq \emptyset$ **do**
  $m \leftarrow \arg\max_{m \in A} \left[F(S \cup \{m\}) - F(S)\right] / c(m)^{\varrho}$; remove m from A
  if the gain is 0 **then break**
  end if
  if $c(S) + c(m) \le B$ **then** $S \leftarrow S \cup \{m\}$
  end if
end while
return S

ter n_k observed changes over E_k item-days of exposure, the posterior predictive probability that an item confirmed a days ago still holds is

$$\hat{\rho}_k(a) = \left(1 + \frac{a}{\beta + E_k}\right)^{-(\alpha + n_k)}. \tag{4}$$

The class hazards are estimated from the store's own version history, so stable classes stop decaying without any tuning. (4) is a Lomax tail, which is a power law for ages well beyond $\beta + E_k$; at smaller ages it behaves like an exponential. So the mixture argument behind this estimator gives the same tail as the power-law decay seen in human memory and in the environment [2]; see Proposition 10.

Keyed facts have exact version chains: a superseded item is dropped from the candidates, and a query with historical intent ("previously", "used to") brings old versions back with $\rho = 1$. Text without keys has no such chain. For it we use a heuristic: a newer item with similarity above 0.8 discounts an older one (the product in Algorithm 1), and a newer near-twin above 0.97 removes it. Without embeddings, the similarity comes from feature hashing. This rule fires only on near-identical text, so it suits structured notes and rarely fires on natural-language updates; Section 6 measures both cases.

4.4 Learning

The online layer sits on top of the frozen relevance model. It keeps a ridge estimate over three features, a bias, the model's probability and the item's past use, starting from $\theta_0 = (0, 1, 0)$, so that with no feedback it returns the model's probability: $\hat{\theta}_t = \theta_0 + V_t^{-1} \sum_{(x,y)} (y - x^\top \theta_0) x$ with $V_t = \lambda I + \sum x x^\top$. The bias is used in fitting and zeroed when scoring. It absorbs the base rate of the reward, including false positives from a noisy signal; a constant added to every item would otherwise push the density rule toward short items. Feedback $y \in [0, 1]$ says whether an injected item was used, and a used item counts as reconfirmed. Items known to be invalid at feedback time are skipped, since they say nothing about μ. The update is $O(d^2)$ by Sherman–Morrison per item and works offline. In the plugin, an injected item counts as used when its identifiers or paths appear in later tool calls or edits, and a failed test halves the reward, following the evidence that self-reports inflate reward [5]. Item-level counters always update, but policy learning is opt-in, because Section 6 finds it fragile under noise.

4.5 Consolidation and forgetting

Consolidation is reversible. Star clusters around a leader, with every member above a similarity threshold of 0.9 by default, are summarized extractively: density greedy on a sentence-level facility location with rate R, which yields the rate–distortion guarantee of Theorem 3. The summary is a new semantic item with parent links, the sources move to an archive tier, and EXPAND restores them on demand. This follows the evidence that repeated irreversible rewriting erodes detail [13, 71]. Forgetting moves items to the archive when they have been idle for τ days or when the active store exceeds a token capacity, evicting the lowest $\rho \cdot \text{activation} \cdot \text{usefulness}/c$ first. Both rules are opt-in, because Section 6 finds that they cost recall. Deletion happens only on an explicit user command.

4.6 Multi-resolution

An item above 160 tokens can also enter selection as an excerpt: the run of consecutive sentences within 128 tokens that covers the most *idf*-weighted query terms, marked with the item's id. The excerpt's weight is the item's probability times the share of query weight it keeps, or its share of the tokens if that is larger, and its cost is its own. The two versions of an item are mutually exclusive. That adds a partition constraint to the knapsack, which Theorem 1 does not cover, so excerpts are a heuristic. The agent pulls the full text with EXPAND when it needs it. This mode is off by default.

4.7 Complexity

With N active items, n candidates, embedding dimension D, $\mathcal{T}$ trees of depth at most h, and $|S| = k$ selected: a write costs $O(\log N)$ plus one embedding; candidate generation costs $O(ND)$ by brute force (or sub-linear with an ANN index) plus the FTS5 query; scoring costs $O(n\mathcal{T}h)$; on the modular objective, greedy reduces to a sort and costs $O(n \log n)$ per run. A bandit update costs $O(kd^2)$. Space is $O(N(D + \bar{c}))$. The memory layer itself uses no LLM tokens; the only token cost is the injected $c(S) \le B$.

5 Theory

All proofs are in Appendix A. Throughout, F is any set function with the properties of Lemma 1.

5.1 Selection

Theorem 1 (Selection guarantee)**.** *Let F be non-negative monotone submodular with $F(\emptyset) = 0$ and let $c(m) \le B$ for all candidates. The set returned by Algorithm 1 satisfies $F(S) \ge \frac{1}{2}(1 - \frac{1}{e})\,\mathrm{OPT}$.*

This is the classical cost-benefit argument [30, 34], restated with a full proof because the constant in related analyses has needed correction [58]. Better constants exist for this algorithm family, 0.405 [58], and $1 - 1/e$ with partial enumeration at higher cost [17, 56]. The worst case is pessimistic for a practical reason: at each query the following bound can be computed.

Proposition 6 (Per-query certificate)**.** *For any feasible S, with $\Delta(m \mid S) = F(S \cup \{m\}) - F(S)$,*

$$\mathrm{OPT} \le F(S) + \max_{\substack{y \in [0,1]^M \\ \sum_m y_m c(m) \le B}} \sum_{m \notin S} y_m \, \Delta(m \mid S), \quad (5)$$

and the right-hand side is computed by sorting on Δ/c in $O(n \log n)$.

5.2 Selection with estimated probabilities

With the relevance model of Section 4, the objective is modular: $F(S) = \sum_{m \in S} w_m$, the expected number of needed items in S when w_m is the probability that m is needed. The model only estimates these probabilities, so the question is how estimation error reaches the selection.

Proposition 7 (From probability error to selection loss)**.** *Let p_m be the probability that candidate m is needed and $\hat{p}_m \ge 0$ an estimate, and write $e(S) = \sum_{m \in S} |p_m - \hat{p}_m|$. Let $S^\star$ maximise $\sum_{m \in S} p_m$ under the budget. If $\hat{S}$ is feasible and $\sum_{m \in \hat{S}} \hat{p}_m \ge \alpha \max_{c(S) \le B} \sum_{m \in S} \hat{p}_m$, then*

$$\sum_{m \in \hat{S}} p_m \;\ge\; \alpha \sum_{m \in S^\star} p_m - \alpha\, e(S^\star) - e(\hat{S}).$$

Density greedy with the best singleton has $\alpha = \frac{1}{2}$ on modular objectives, and an exact knapsack has $\alpha = 1$, in which case the loss is at most $e(S^\star \triangle \hat{S})$.

So only the error on selected and optimal items matters, not the error on the whole pool. The worst case separates the two algorithms by a factor of two, while on our dev data they differ by about a point, which is why the simpler one is kept.

5.3 Learning with expiring items

The closest results [57, 68, 70] assume that an item's expected value is a stationary linear function of its features. Memory violates this: a fact's value collapses when the fact stops holding, while its text, and so its features, stay the same. We therefore model the two factors separately.

Setting. In round $t = 1, \dots, T$ the learner sees candidates C_t with features $\|x_{t,m}\|_2 \le 1$ (rescaled by a constant if needed), costs, and a function $F_t(S; w)$ satisfying Lemma 1 with constant L. There is an unknown $\theta^\star$ with $\|\theta^\star - \theta_0\|_2 \le \Sigma$ and $\mu_{t,m} = x_{t,m}^\top \theta^\star \in [0,1]$. Each item is valid with probability $\rho_{t,m} \in [0,1]$ given the history before round t, which may vary arbitrarily with t, and the true weight is $w^\star_{t,m} = \rho_{t,m}\mu_{t,m}$. For each injected item the learner observes $Z_{t,m} \in \{0,1\}$ with mean $\rho_{t,m}$, possibly correlated within a round, and, if $Z_{t,m} = 1$, a reward $y_{t,m} = \mu_{t,m} + \xi_{t,m}$, where $\xi_{t,m}$ is R-sub-Gaussian given the past, the current Z and the earlier rewards of the round. At most K items are injected per round. A validity estimate satisfies $|\hat{\rho}_{t,m} - \rho_{t,m}| \le \varepsilon_{t,m}$ for all t, m with probability at least $1 - \delta/2$.

Algorithm. Use $\bar{w}_{t,m} = \min(1, \hat{\rho}_{t,m} + \varepsilon_{t,m}) \cdot \mathrm{clip}_{[0,1]}(x_{t,m}^\top \hat{\theta}_t + \beta_t \|x_{t,m}\|_{V_t^{-1}})$, with $\hat{\theta}_t, V_t$ the prior-centred ridge estimates over the rounds where $Z = 1$, and pick S_t with an α-approximate oracle for $F_t(\cdot\,; \bar{w}_t)$ under the budget.

Theorem 2 (Validity-aware regret)**.** *Let $\lambda \ge K$ and $\beta_t = \beta_T := R\sqrt{d \log\big(4(1 + KT/\lambda)/\delta\big)} + \sqrt{\lambda}\,\Sigma$. With probability at least $1 - \delta$,*

$$\sum_{t=1}^{T} \Big[\alpha \max_{c(S) \le B} F_t(S; w_t^\star) - F_t(S_t; w_t^\star)\Big] \le L\left[2\beta_T \sqrt{2dKT \log\left(1 + \frac{KT}{d\lambda}\right)} + K\sqrt{2T \log\frac{4}{\delta}} + 2\sum_{t=1}^{T} \sum_{m \in S_t} \varepsilon_{t,m}\right].$$

The comparator is the per-round optimum under the time-varying $w_t^\star$, so the bound is dynamic, and the $\rho_{t,m}$ need no variation budget: the burden moves into estimating them, through the last term. The first term is the usual linear bandit price, the second comes from the randomness of validity, and warm starting helps through Σ. With $\alpha = \frac{1}{2}(1 - \frac{1}{e})$ from Theorem 1 and $L = (1-\gamma)(1+\eta)$, the theorem covers Algorithm 1 run with β_T and the margin ε.

What this does and does not say. The theorem describes how regret scales, and it is not a practical

guarantee. Since $F \le 1$ in the original objective, regret is trivially at most T, and with $K = 40$, $R = \frac{1}{2}$, $\Sigma = 1$, $\delta = 0.05$ and $L = 1$, the bound falls below T near 1.2×10^7 rounds at $d = 12$. Letting the online layer see only three features, with the frozen model's probability as one of them, moves that point to 1.6×10^6 rounds, which is better and still far from a session count. The shipped defaults use $\beta = 0$ and no margin, and they use a heuristic rule on top of (4), so the code is not the analyzed algorithm. The recall tables in Section 6 use the static model and do not rest on online learning at all.

Proposition 8 (Validity error of the hazard estimate)**.** *Let items of class k have lifetimes that are* $\mathrm{Exp}(\lambda_k)$*, with right-censoring that depends only on the past. Let N be the number of observed changes, E the total exposure up to a stopping time, and $\hat{\lambda}_k = N/E$. If $E \ge E_0$ almost surely and $\eta := 2\sqrt{\log(2/\delta)/(\lambda_k E_0)} \le \frac{1}{2}$, then with probability at least $1 - \delta$, for all ages $a \ge 0$ simultaneously,*

$$|e^{-\hat{\lambda}_k a} - e^{-\lambda_k a}| \le \frac{2}{e}\,\eta.$$

The bound does not depend on age: the error is governed by $\lambda_k E_0$, the expected number of observed changes. This estimator counts censored exposure, as the code does, whose posterior (4) has the same limit. If every class in use accumulates exposure as $\lambda_k E_0 \ge \nu t$, then $\varepsilon_{t,m} = \tilde{O}(1/\sqrt{\nu t})$ by a union bound over a grid of times, $\sum_t \sum_{m \in S_t} \varepsilon_{t,m} = \tilde{O}(K\sqrt{T/\nu})$, and the total is $\tilde{O}((d\sqrt{K} + K/\sqrt{\nu})\sqrt{T})$. For the conditional probabilities of the setting to match this survival curve, age must restart when an item is confirmed; the code does this for items that are used.

Proposition 9 (Validity-blind learners)**.** *Let $m \ge 2$. There is an instance in which m items per round have identical features, exactly one is valid, and B fits one item. Any learner that does not observe age or version information has $\mathbb{E}[\sum_t \alpha\,\mathrm{OPT}_t - F_t(S_t)] \ge (\alpha - \frac{1}{m})T$ for every $\alpha > 1/m$. So it has linear regret, for all horizons, whereas Theorem 2 applies with $\varepsilon \equiv 0$ if the version chain is known.*

5.4 Validity and activation

Proposition 10 (Power-law survival from a hazard mixture)**.** *If hazards in a class are* $\mathrm{Gamma}(\alpha, \beta)$ *distributed, the population survival is $S(a) = (1 + a/\beta)^{-\alpha}$. As $a \to \infty$, $\log S(a) = -\alpha \log a + O(1)$, the form of ACT-R single-trace base-level activation with decay $d = \alpha$.*

5.5 Consolidation

Let X be a cluster of sentences with weights p ($\sum_x p_x = 1$), costs c, cosine similarities $s \in [0,1]$ with $s(x,x) = 1$, and let $\Phi(Z) = \sum_x p_x \max_{z \in Z} s(x,z)$, $D(Z) = 1 - \Phi(Z)$ be the distortion of a summary $Z \subseteq X$.

Theorem 3 (Rate–distortion for extractive consolidation)**.** *Let Z_ℓ be the first ℓ picks of density greedy on Φ, with rate $R_\ell = c(Z_\ell)$. For every summary $Z^\star$ of rate $R^\star$,*

$$D(Z_\ell) \le e^{-R_\ell/R^\star} + \big(1 - e^{-R_\ell/R^\star}\big) D(Z^\star).$$

Moreover, for every query vector e_q and every ℓ_{u}-Lipschitz item score $u_q(x) = \phi(\langle e_q, e_x \rangle)$,

$$\mathbb{E}_{x \sim p}\big[u_q(x) - u_q(z(x))\big] \le \ell_{\mathrm{u}} \sqrt{2\,D(Z_\ell)},$$
$$z(x) = \arg\max_{z \in Z_\ell} s(x,z).$$

The token saving is $c(X) - R_\ell$, so distortion decays exponentially in rate against any reference summary. The statement covers the greedy prefix; the implementation continues past the first sentence that does not fit, as a heuristic. The bound averages over p and says nothing about a rare needle in the cluster. That is why consolidation here keeps the sources: if the summary is retrieved, EXPAND recovers any source at the cost of its tokens.

5.6 Store size

Theorem 4 (Store stability)**.** *Let items arrive as a Poisson process of rate a, and let each be accessed by an independent Poisson process of rate r, drawn i.i.d. from a distribution with $\mathbb{E}[(e^{r\tau} - 1)/r] < \infty$. An item is archived after an idle gap of τ. Then an item's active lifetime has mean $(e^{r\tau} - 1)/r$ given r, and the number of active items converges in distribution to* $\mathrm{Poisson}\big(a\,\mathbb{E}[(e^{r\tau} - 1)/r]\big)$.

Without forgetting, the store grows linearly in t. With an idle rule, the active store is bounded in distribution, and the retained set is biased exponentially toward frequently used items. The theorem says nothing about what forgetting costs in recall, and it assumes independent access processes, whereas real access patterns are bursty. Section 6 measures both.

6 Experiments

6.1 Setup

Mode. All metrics are retrieval-level: no LLM takes part in the method or in the evaluation, so we report what reaches the context and not what a model does with it.

Development data. *LongMemEval-S* [65] has 500 questions, each with its own store of about 490 turns. *LoCoMo* [37] has 10 long conversations. *BEAM-1M*

[60] has 35 long conversations of about 2,000 turns. *RepoMem* (new) turns eight Python repositories into memory streams, at 14 snapshots over six years. Each interval is a session in which a simulated agent records commit notes and facts observed in changed files. Facts (constants, signatures, symbol locations, dependency constraints) come from `ast`, so a fact is stale exactly when the same key held another value at an earlier snapshot. Fact queries ask for the current value; task queries use a real later commit subject, with earlier commits on the same files as gold. RepoMem runs with slot keys, as a host agent would supply, and without.

Held-out test. *BEAM-100K and BEAM-500K*: 984 questions in 55 conversations that share no seed with BEAM-1M, keeping every question with labelled evidence. Their turns average 232 tokens for evidence and 482 for the rest, and the full history of a conversation averages 379,187 tokens. *RepoMem* on five repositories outside development (marshmallow, pluggy, packaging, loguru, tqdm): 1,248 queries over three seeded streams each. No model, threshold or design choice was made on this data. The method was frozen first, and the test ran once, with a single script.

Baselines. Recency, BM25 (FTS5), dense (all-MiniLM-L6-v2 [61]), hybrid RRF [14], hybrid followed by a cross-encoder (ms-marco-MiniLM-L-6-v2) on the top 50, MMR [8], Generative Agents scoring [45] with constant importance, facility location under a knapsack as in PACMS [19], hybrid with superseded versions removed (temporal invalidation alone, in the spirit of [49]), hybrid listed newest first, and hybrid keeping only the newest of near-twins. We also keep the *first version* of the method: a linear prior over retrieval signals with no item, role or context features, and a pool of 100 from each retriever. All methods share candidate generators, token counts (`cl100k_base`) and a 14-token per-item overhead, and every ranking is filled greedily to the budget, which already makes the baselines budget-aware. Systems that need an LLM on write were not run.

Statistics. The independent unit is a conversation on BEAM (55) and a repository on RepoMem (5), since queries within a conversation are dependent and seeds repeat RepoMem queries. We give bootstrap intervals on the unit, and p-values from two-sided sign-flip permutation tests on the unit means, with Holm correction across budgets within a comparison. With five units the smallest possible two-sided p is 1/16, so no RepoMem comparison can reach significance. There, read the interval and the count of units favouring each side, and note that an interval containing zero on five units is weak evidence that two methods are equal.

6.2 Development results

Table 2 gives cross-fitted recall: each query is scored by a model that never saw its question, conversation or repository. The relevance model raises mean recall over the first version on all four datasets, from 0.751 to 0.801 on LongMemEval-S, 0.663 to 0.732 on LoCoMo, 0.863 to 0.881 on RepoMem and 0.344 to 0.457 on BEAM-1M. The large-budget loss to hybrid ranking disappears on LoCoMo: 0.920 against 0.861 at 4096 tokens. Without a cross-encoder the method still trails the reranker at the smallest budget, 0.454 against 0.517 on LoCoMo at 128 tokens, because at that size one item decides everything and only joint reading of query and text separates the top candidates. With cross-encoder features it closes that gap, at 0.530.

The candidate pool matters most on BEAM, where a question has many evidence turns. There the top 100 from each retriever reach 0.665 of the evidence, and 200 reach 0.744, at a cost of 326 candidates instead of 167. Adding the neighbours of the 50 best hits reaches 0.731 with 225 candidates, so neighbours add about twice as much evidence per candidate as a deeper pool, while pseudo-relevance feedback adds little (0.677).

Table 3 removes one feature group at a time and refits. Item features and neighbourhood features carry most of the gain, and query–item overlap adds little on top of them.

The cross-fitted probabilities are close to calibrated: over ten equal-count bins the expected calibration error is 0.0005 on LongMemEval-S, 0.0003 on LoCoMo, 0.0016 on RepoMem and 0.0017 on BEAM-1M, against base rates near one percent. They are somewhat overconfident at the top on BEAM-1M, where the highest-scored one percent of candidates are predicted at 0.349 and observed at 0.287.

With such weights, the choice of knapsack solver hardly matters: over the four datasets, mean recall is 0.801 for density greedy against 0.800 for an exact knapsack on LongMemEval-S, and 0.457 against 0.454 on BEAM-1M. Cost still matters where item lengths vary: ranking by probability alone falls to 0.431 on BEAM-1M. We keep density greedy because Theorem 1 covers it.

6.3 Held-out test

Figure 1 and Table 4 give the results, and Table 5 the paired differences.

Long conversations. On BEAM the method pulls clearly ahead of every rival from 1024 tokens, and the lead grows with the budget. Averaged over budgets it has 0.566 against 0.478 for the reranker and 0.440 for hybrid, a difference of +0.088 over the reranker with unit interval [+0.072, +0.102], and 52/55 conversations favour it. At 1024 tokens recall is 0.635 against 0.488.

Table 2: Cross-fitted evidence recall on development data. Mean is over the six budgets.

Method	128	256	512	1024	2048	4096	mean
LongMemEval-S							
First version	0.406	0.641	0.770	0.853	0.903	0.935	0.751
Hybrid RRF	0.379	0.593	0.669	0.756	0.848	0.907	0.692
Hybrid + reranker	0.442	0.695	0.783	0.860	0.906	0.933	0.770
BM25	0.392	0.611	0.675	0.743	0.811	0.861	0.682
KANERVA	0.407	0.673	0.845	0.926	0.970	0.986	0.801
KANERVA + reranker	0.434	0.700	0.866	0.931	0.970	0.985	0.814
KANERVA, lexical	0.383	0.661	0.814	0.893	0.928	0.941	0.770
KANERVA, static embedder	0.402	0.684	0.845	0.910	0.964	0.986	0.799
LoCoMo							
First version	0.408	0.555	0.661	0.733	0.785	0.837	0.663
Hybrid RRF	0.335	0.476	0.592	0.690	0.788	0.861	0.624
Hybrid + reranker	0.517	0.631	0.696	0.733	0.791	0.861	0.705
BM25	0.410	0.548	0.630	0.699	0.753	0.799	0.640
KANERVA	0.454	0.617	0.722	0.808	0.870	0.920	0.732
KANERVA + reranker	0.530	0.662	0.758	0.824	0.873	0.924	0.762
KANERVA, lexical	0.441	0.568	0.654	0.729	0.776	0.813	0.663
KANERVA, static embedder	0.462	0.611	0.716	0.797	0.861	0.907	0.725
RepoMem							
First version	0.771	0.824	0.858	0.881	0.903	0.940	0.863
Hybrid RRF	0.777	0.818	0.859	0.886	0.921	0.946	0.868
Hybrid + reranker	0.808	0.851	0.875	0.894	0.922	0.946	0.882
BM25	0.717	0.767	0.818	0.857	0.885	0.907	0.825
KANERVA	0.802	0.839	0.871	0.897	0.926	0.954	0.881
KANERVA + reranker	0.812	0.847	0.875	0.901	0.930	0.956	0.887
KANERVA, lexical	0.722	0.776	0.840	0.875	0.897	0.910	0.837
KANERVA, static embedder	0.722	0.782	0.833	0.872	0.919	0.949	0.846
BEAM-1M							
First version	0.193	0.288	0.363	0.364	0.404	0.450	0.344
Hybrid RRF	0.198	0.303	0.372	0.309	0.340	0.368	0.315
Hybrid + reranker	0.218	0.319	0.391	0.351	0.385	0.425	0.348
BM25	0.165	0.256	0.291	0.241	0.263	0.296	0.252
KANERVA	0.212	0.333	0.438	0.518	0.593	0.651	0.457
KANERVA + reranker	0.217	0.347	0.442	0.524	0.592	0.652	0.462
KANERVA, lexical	0.181	0.276	0.352	0.411	0.448	0.476	0.357
KANERVA, static embedder	0.185	0.295	0.388	0.481	0.543	0.605	0.416

Table 3: Mean recall lost when a feature group is removed and the model refitted, on development data. Positive means the group helps.

Group removed	LongMemEval-S	LoCoMo	RepoMem	BEAM-1M
Item	+0.008	+0.032	+0.010	+0.008
Role and speaker	+0.003	+0.011	-0.000	+0.010
Query–item overlap	+0.001	-0.002	+0.001	+0.005
Neighbourhood	+0.008	+0.033	+0.001	+0.009
Recency and activation	+0.001	-0.004	+0.002	+0.010

Below 1024 tokens the difference from the reranker is not significant, and at 128 tokens the two have 0.324 and 0.322. Adding cross-encoder features gives 0.350 at 128 tokens and 0.644 at 1024. The gain over the first version is +0.086 with interval [+0.074, +0.098]. Removing the wide pool and neighbours costs +0.016, and removing cost density costs +0.016 (Table 6).

Code history. On RepoMem the picture is mixed. The method leads at the smallest budget, with keys 0.782 against 0.726 for the reranker and 0.764 for hybrid with invalidation. From 1024 tokens it falls behind: at 4096 tokens with keys it has 0.926 against 0.952, and without keys 0.917 against 0.951. Averaged over budgets the difference to hybrid is +0.007 with keys and -0.009 without, with intervals [-0.024, +0.028] and [-0.041, +0.017] that contain zero on five repositories. The first version is beaten on every repository, by +0.015 without keys. Most of the loss at large budgets sits on the task track, and it comes from the near-twin rules, which take look-alike commit notes for versions of one fact. Without keys at 4096 tokens, task recall is 0.642, against 0.788 with those rules removed and 0.761 for hybrid, while fact recall is 0.988 against 1.000, which accounts for the rest. Removing the rules altogether would bring back the stale injections shown next, so restricting them to facts is the obvious change, which we did not test.

Lighter configurations. The lexical model needs no embeddings and is the plugin default. On BEAM it has 0.488 against 0.374 for BM25, on the same candidates. On RepoMem without keys it has 0.809 against 0.818, and it loses to BM25 at 4096 tokens, 0.895 against 0.939. With the static model2vec embedder [62], which needs no neural runtime, the method reaches 0.536 on BEAM, better than the lexical model, but 0.779 on RepoMem, worse than the lexical model and worse than BM25. On changed RepoMem facts it also injects a stale version more often, 0.371 against 0.206. So the configurations the plugins can run are strong on conversations and not on code facts: the numbers in the first paragraphs of this section need the MiniLM embedder.

Mem0. On a random 40-question development sample of LongMemEval-S, Mem0's verbatim mode reached 0.829 at 1024 tokens against 0.877 for the first version of the method and 0.846 for hybrid on the same questions, with 9 ms per write. Mem0's LLM extraction, its main feature, was not exercised.

6.4 Staleness

Table 7 reports, on RepoMem queries whose fact changed, how often an outdated version is *injected*, and the *error*: how often it is injected and the current version is absent or listed below it. Similarity retrieval injects the outdated version nearly always, 0.997 of the time at 1024 tokens. With keys, exact version chains remove stale items entirely, and hybrid with invalidation does this as well as KANERVA. Without keys, KANERVA injects a stale version 0.194 of the time, and the simple rule of keeping the newest near-twin reaches 0.227. So on RepoMem the gain comes from that rule, which KANERVA also uses, and not from its probabilistic validity model. We have no ablation that isolates the hazard model's value.

The rule depends on near-identical text, and RepoMem's versions differ only in a value. On the *natural-language* updates of LongMemEval, evidence for a fact and for its update are not near-twins, the rule does not fire, and on development data KANERVA injects the outdated turn on 1.000 of questions against 0.857 for hybrid, since it finds more evidence of both kinds. Listing items newest first keeps the error to 0.029, against 0.471 for hybrid in rank order. Detecting that a free-text statement replaces an older one remains unsolved

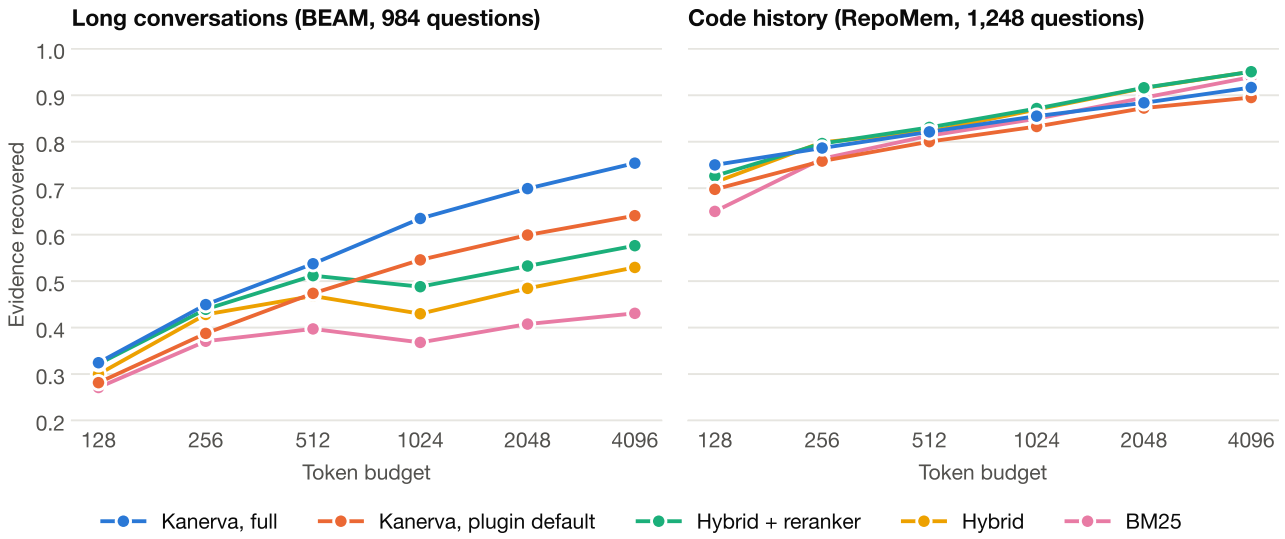


Figure 1: Evidence recall at each token budget on held-out data. RepoMem is shown without keys.

here, and it likely needs semantics that an LLM-free method lacks.

6.5 Excerpts

Turn recall credits an excerpt that lost the answer, so excerpts are compared on *answer-term recall*: the share of the reference answer's content stems, excluding those in the question, found in the injected text. On BEAM at 128, 256 and 512 tokens, excerpts of long items raise it from 0.280, 0.384 and 0.475 to 0.289, 0.395 and 0.489. The differences are small, +0.010 to +0.013, and they fade by 1024 tokens, so the mode stays off by default.

6.6 Online learning

Feedback here is simulated from evidence labels, as described in Section 7. Each seed shuffles the order of work and shares a single policy. Table 8 reports recall at $B = 512$. On development data, where the model already fits well, online learning does not help. On LoCoMo recall moves from 0.746 to 0.725, and on RepoMem from 0.899 to 0.896, and LongMemEval-S gives each question its own store, so nothing carries over. Past use of an item for another query turns out to be a poor predictor for the next one. With 20% of rewards flipped, recall on LoCoMo is 0.729. So policy learning stays opt-in.

6.7 Checks of the theory on synthetic instances

With known ground truth (eight seeds, horizon 4000, 40 items per round, three classes with different hazards), cumulative regret against the per-round greedy optimum is 22 for a learner that knows validity, 26 when it estimates hazards from censored lifetimes, and 2015 when it is blind. The log–log slopes in T are 0.26, 0.25 and 0.97. The blind learner's linear growth is what Proposition 9 predicts, and it shows the value of the separation. The observed regret of the other two is far below the upper bound of Theorem 2, whose value here exceeds the trivial bound, so these runs do not test the bound's tightness. Over 1600 random selection problems, $F(S)$ was on average 0.707 of the certificate upper bound, with a minimum of 0.600, above the worst-case constant of 0.316.

6.8 Consolidation and forgetting

Consolidation. Table 9 applies consolidation at several thresholds. A gold item counts as found if it or its summary is selected, which measures discoverability, since the sources stay expandable. Consolidation is not free. At $\tau = 0.9$ it merges only near-duplicates: it saves 2.1% of tokens on the LongMemEval-S sample and 0.5% on LoCoMo, with recall unchanged. At $\tau = 0.8$ it saves 19.9% and 5.6%, and recall moves from 0.969 to 0.789 and from 0.824 to 0.791. At $\tau = 0.7$ the loss is larger. Since the budget caps injected tokens anyway, a smaller store buys little here, so the default threshold is 0.9.

Forgetting. On RepoMem streams, Table 10 compares keeping everything with idle-time rules and token capacities. Forgetting has a real cost on this benchmark. Idle rules of 365 and 180 days cut the active store to 6176 and 3227 tokens from 21492, as Theorem 4 leads one to expect, but recall falls from 0.923 to 0.713 and 0.551, because long-lived facts are read rarely and are

still correct. Sessions in this benchmark are 168 days apart, so these idle rules are short in session terms, and the numbers do not transfer directly to a daily workflow. Token capacities behave similarly. Given the latencies below, a store of this size does not need forgetting, so the rules are opt-in. Archived items remain recoverable by hand, but they leave retrieval.

6.9 Latency and scale

On an M1 Pro with an uncontended CPU, a fresh query at 10^4 items takes 287.1 ms at the median and 359.7 ms at the 95th percentile, plus 8 ms to embed the query, and lexical retrieval takes 150.2 and 210.5 ms. At 10^5 items the medians are 332.4 and 182.2 ms. A write costs 0.49 ms, excluding the embedding. The memory layer spends no LLM tokens, so its token cost is the injected context alone.

Table 4: Evidence recall on held-out data. Bold marks the best method per column within a dataset.

Method	128	256	512	1024	2048	4096	mean
BEAM-100K and 500K							
KANERVA + reranker	**0.350**	**0.480**	**0.571**	**0.644**	**0.707**	0.753	**0.584**
KANERVA	0.324	0.449	0.537	0.635	0.699	**0.754**	0.566
KANERVA, lexical	0.281	0.388	0.474	0.546	0.599	0.641	0.488
First version	0.312	0.434	0.494	0.493	0.549	0.600	0.480
Hybrid + reranker	0.322	0.439	0.512	0.488	0.533	0.576	0.478
Hybrid RRF	0.300	0.428	0.468	0.430	0.484	0.529	0.440
Hybrid, live only	0.300	0.428	0.468	0.430	0.484	0.529	0.440
Hybrid, newest of twins	0.300	0.428	0.467	0.430	0.485	0.528	0.440
Dense	0.272	0.380	0.458	0.450	0.506	0.553	0.436
BM25	0.271	0.371	0.397	0.368	0.408	0.431	0.374
MMR	0.257	0.356	0.446	0.442	0.504	0.562	0.428
Generative Agents	0.012	0.019	0.024	0.021	0.024	0.028	0.021
Facility location	0.065	0.125	0.222	0.369	0.538	0.668	0.331
Recency	0.006	0.008	0.010	0.008	0.012	0.013	0.009
RepoMem, keys							
KANERVA + reranker	**0.782**	**0.809**	**0.835**	0.866	0.895	0.926	0.852
KANERVA	0.782	0.805	**0.835**	0.869	0.896	0.926	0.852
KANERVA, lexical	0.721	0.780	0.817	0.847	0.884	0.904	0.826
First version	0.759	0.804	0.827	0.857	0.883	0.915	0.841
Hybrid + reranker	0.726	0.796	0.831	0.871	0.916	0.951	0.849
Hybrid RRF	0.712	0.799	0.822	0.869	0.915	0.951	0.845
Hybrid, live only	0.764	0.802	0.833	**0.880**	**0.919**	**0.952**	**0.858**
Hybrid, newest of twins	0.756	0.799	0.830	0.874	0.908	0.927	0.849
Dense	0.734	0.798	0.830	0.868	0.906	0.940	0.846
BM25	0.650	0.763	0.813	0.850	0.893	0.939	0.818
MMR	0.687	0.784	0.821	0.866	0.909	0.945	0.835
Generative Agents	0.211	0.381	0.659	0.825	0.900	0.940	0.653
Facility location	0.103	0.168	0.251	0.378	0.549	0.791	0.374
Recency	0.024	0.048	0.077	0.116	0.210	0.370	0.141
RepoMem, no keys							
KANERVA + reranker	0.744	0.786	0.819	0.852	0.883	0.916	0.833
KANERVA	0.750	0.786	0.821	0.855	0.884	0.917	0.836
KANERVA, lexical	0.698	0.758	0.800	0.833	0.872	0.895	0.809
First version	0.713	0.776	0.809	0.845	0.871	0.906	0.820
Hybrid + reranker	0.726	0.796	**0.831**	**0.871**	**0.916**	**0.951**	**0.849**
Hybrid RRF	0.713	**0.799**	0.822	0.869	0.915	**0.951**	0.845
Hybrid, live only	0.713	**0.799**	0.822	0.869	0.915	**0.951**	0.845
Hybrid, newest of twins	**0.754**	0.798	0.826	0.870	0.904	0.923	0.846
Dense	0.734	0.798	0.830	0.868	0.906	0.940	0.846
BM25	0.650	0.763	0.813	0.850	0.895	0.939	0.818
MMR	0.688	0.785	0.821	0.866	0.909	0.945	0.836
Generative Agents	0.211	0.380	0.657	0.825	0.900	0.940	0.652
Facility location	0.103	0.168	0.252	0.380	0.550	0.792	0.374
Recency	0.024	0.048	0.076	0.113	0.209	0.371	0.140

Table 5: Paired differences in recall on held-out data. Positive favours the first method named. Unit CI resamples conversations or repositories, Units + counts those favouring the first method, and p is Holm-corrected across budgets, except on the mean rows.

Dataset	Comparison	B	Δ recall	Unit CI	Units +	p
BEAM-100K and 500K	KANERVA vs hybrid	128	+0.025	[+0.010, +0.038]	34/55	0.001
BEAM-100K and 500K	KANERVA vs hybrid	1024	+0.205	[+0.184, +0.227]	55/55	<0.001
BEAM-100K and 500K	KANERVA vs hybrid	4096	+0.224	[+0.203, +0.246]	55/55	<0.001
BEAM-100K and 500K	KANERVA vs hybrid	mean	+0.127	[+0.114, +0.140]	55/55	<0.001
BEAM-100K and 500K	KANERVA vs reranker	128	+0.003	[-0.013, +0.018]	29/55	0.739
BEAM-100K and 500K	KANERVA vs reranker	1024	+0.147	[+0.121, +0.172]	51/55	<0.001
BEAM-100K and 500K	KANERVA vs reranker	4096	+0.178	[+0.156, +0.200]	55/55	<0.001
BEAM-100K and 500K	KANERVA vs reranker	mean	+0.088	[+0.072, +0.102]	52/55	<0.001
BEAM-100K and 500K	KANERVA vs first version	128	+0.012	[-0.001, +0.025]	33/55	0.128
BEAM-100K and 500K	KANERVA vs first version	1024	+0.142	[+0.120, +0.164]	52/55	<0.001
BEAM-100K and 500K	KANERVA vs first version	4096	+0.154	[+0.136, +0.174]	55/55	<0.001
BEAM-100K and 500K	KANERVA vs first version	mean	+0.086	[+0.074, +0.098]	53/55	<0.001
BEAM-100K and 500K	+ reranker vs reranker	128	+0.028	[+0.015, +0.041]	41/55	<0.001
BEAM-100K and 500K	+ reranker vs reranker	1024	+0.156	[+0.133, +0.178]	53/55	<0.001
BEAM-100K and 500K	+ reranker vs reranker	4096	+0.177	[+0.155, +0.198]	54/55	<0.001
BEAM-100K and 500K	+ reranker vs reranker	mean	+0.106	[+0.092, +0.119]	54/55	<0.001
BEAM-100K and 500K	Lexical vs BM25	128	+0.010	[-0.002, +0.023]	28/55	0.125
BEAM-100K and 500K	Lexical vs BM25	1024	+0.178	[+0.150, +0.203]	54/55	<0.001
BEAM-100K and 500K	Lexical vs BM25	4096	+0.210	[+0.186, +0.234]	55/55	<0.001
BEAM-100K and 500K	Lexical vs BM25	mean	+0.114	[+0.101, +0.128]	55/55	<0.001
RepoMem, keys	KANERVA vs hybrid	128	+0.070	[+0.027, +0.125]	5/5	0.375
RepoMem, keys	KANERVA vs hybrid	1024	-0.000	[-0.036, +0.021]	4/5	1.000
RepoMem, keys	KANERVA vs hybrid	4096	-0.025	[-0.069, +0.012]	3/5	1.000
RepoMem, keys	KANERVA vs hybrid	mean	+0.007	[-0.024, +0.028]	4/5	0.625
RepoMem, keys	KANERVA vs reranker	128	+0.055	[+0.026, +0.101]	5/5	0.375
RepoMem, keys	KANERVA vs reranker	1024	-0.003	[-0.031, +0.018]	4/5	1.000
RepoMem, keys	KANERVA vs reranker	4096	-0.025	[-0.069, +0.012]	3/5	1.000
RepoMem, keys	KANERVA vs reranker	mean	+0.004	[-0.022, +0.020]	4/5	0.875
RepoMem, keys	KANERVA vs first version	128	+0.023	[+0.007, +0.040]	4/5	0.750
RepoMem, keys	KANERVA vs first version	1024	+0.012	[+0.001, +0.025]	4/5	0.938
RepoMem, keys	KANERVA vs first version	4096	+0.011	[-0.002, +0.023]	4/5	1.000
RepoMem, keys	KANERVA vs first version	mean	+0.011	[+0.006, +0.017]	5/5	0.062
RepoMem, keys	+ reranker vs reranker	128	+0.056	[+0.030, +0.094]	5/5	0.375
RepoMem, keys	+ reranker vs reranker	1024	-0.006	[-0.033, +0.012]	3/5	1.000
RepoMem, keys	+ reranker vs reranker	4096	-0.025	[-0.069, +0.013]	3/5	1.000
RepoMem, keys	+ reranker vs reranker	mean	+0.004	[-0.023, +0.020]	4/5	0.812
RepoMem, keys	Lexical vs BM25	128	+0.071	[+0.030, +0.126]	5/5	0.375
RepoMem, keys	Lexical vs BM25	1024	-0.002	[-0.043, +0.029]	4/5	1.000
RepoMem, keys	Lexical vs BM25	4096	-0.035	[-0.072, -0.006]	0/5	0.625
RepoMem, keys	Lexical vs BM25	mean	+0.008	[-0.025, +0.035]	4/5	0.562
RepoMem, no keys	KANERVA vs hybrid	128	+0.037	[-0.003, +0.096]	4/5	1.000
RepoMem, no keys	KANERVA vs hybrid	1024	-0.014	[-0.052, +0.011]	3/5	1.000
RepoMem, no keys	KANERVA vs hybrid	4096	-0.034	[-0.082, +0.004]	2/5	1.000
RepoMem, no keys	KANERVA vs hybrid	mean	-0.009	[-0.041, +0.017]	3/5	0.812
RepoMem, no keys	KANERVA vs reranker	128	+0.024	[-0.009, +0.071]	3/5	1.000
RepoMem, no keys	KANERVA vs reranker	1024	-0.016	[-0.049, +0.011]	2/5	1.000
RepoMem, no keys	KANERVA vs reranker	4096	-0.034	[-0.082, +0.004]	2/5	1.000
RepoMem, no keys	KANERVA vs reranker	mean	-0.013	[-0.042, +0.010]	3/5	0.562
RepoMem, no keys	KANERVA vs first version	128	+0.037	[+0.012, +0.069]	4/5	0.625
RepoMem, no keys	KANERVA vs first version	1024	+0.010	[+0.003, +0.018]	5/5	0.375
RepoMem, no keys	KANERVA vs first version	4096	+0.010	[-0.003, +0.023]	4/5	0.625
RepoMem, no keys	KANERVA vs first version	mean	+0.015	[+0.010, +0.025]	5/5	0.062
RepoMem, no keys	+ reranker vs reranker	128	+0.018	[-0.008, +0.057]	4/5	1.000
RepoMem, no keys	+ reranker vs reranker	1024	-0.020	[-0.048, +0.006]	2/5	1.000
RepoMem, no keys	+ reranker vs reranker	4096	-0.035	[-0.083, +0.003]	1/5	1.000
RepoMem, no keys	+ reranker vs reranker	mean	-0.015	[-0.040, +0.007]	2/5	0.438
RepoMem, no keys	Lexical vs BM25	128	+0.047	[+0.005, +0.103]	3/5	1.000
RepoMem, no keys	Lexical vs BM25	1024	-0.017	[-0.056, +0.015]	2/5	1.000
RepoMem, no keys	Lexical vs BM25	4096	-0.044	[-0.086, -0.014]	0/5	0.375
RepoMem, no keys	Lexical vs BM25	mean	-0.009	[-0.043, +0.019]	3/5	0.750

Table 6: Ablations on held-out data: KANERVA minus the variant with a component removed.

Dataset	Comparison	B	Δ recall	Unit CI	Units +	p
BEAM-100K and 500K	Wide pool and neighbours	1024	+0.021	[+0.008, +0.033]	36/55	0.006
BEAM-100K and 500K	Wide pool and neighbours	mean	+0.016	[+0.009, +0.022]	42/55	<0.001
BEAM-100K and 500K	Cost density	1024	+0.016	[+0.006, +0.026]	35/55	0.005
BEAM-100K and 500K	Cost density	mean	+0.016	[+0.012, +0.020]	47/55	<0.001
BEAM-100K and 500K	Validity	1024	+0.000	[-0.007, +0.007]	12/55	0.947
BEAM-100K and 500K	Validity	mean	-0.003	[-0.006, -0.000]	16/55	0.050
BEAM-100K and 500K	Near-twin rules	1024	+0.000	[-0.006, +0.007]	12/55	0.947
BEAM-100K and 500K	Near-twin rules	mean	-0.003	[-0.005, +0.000]	17/55	0.070
RepoMem, keys	Wide pool and neighbours	1024	-0.001	[-0.005, +0.002]	2/5	1.000
RepoMem, keys	Wide pool and neighbours	mean	-0.000	[-0.003, +0.003]	2/5	0.938
RepoMem, keys	Cost density	1024	-0.000	[-0.003, +0.002]	2/5	1.000
RepoMem, keys	Cost density	mean	-0.000	[-0.004, +0.003]	3/5	0.812
RepoMem, keys	Validity	1024	-0.004	[-0.038, +0.021]	3/5	1.000
RepoMem, keys	Validity	mean	+0.008	[-0.023, +0.038]	4/5	0.562
RepoMem, keys	Near-twin rules	1024	-0.009	[-0.039, +0.013]	2/5	1.000
RepoMem, keys	Near-twin rules	mean	-0.007	[-0.030, +0.008]	2/5	0.750
RepoMem, no keys	Wide pool and neighbours	1024	-0.003	[-0.008, +0.001]	1/5	1.000
RepoMem, no keys	Wide pool and neighbours	mean	-0.001	[-0.004, +0.003]	2/5	0.938
RepoMem, no keys	Cost density	1024	+0.000	[-0.005, +0.004]	3/5	1.000
RepoMem, no keys	Cost density	mean	+0.001	[-0.002, +0.004]	3/5	0.750
RepoMem, no keys	Validity	1024	-0.019	[-0.056, +0.010]	1/5	1.000
RepoMem, no keys	Validity	mean	-0.008	[-0.041, +0.023]	2/5	0.625
RepoMem, no keys	Near-twin rules	1024	-0.019	[-0.055, +0.010]	1/5	1.000
RepoMem, no keys	Near-twin rules	mean	-0.008	[-0.040, +0.022]	2/5	0.625

Table 7: Staleness on held-out RepoMem queries whose fact changed, at $B = 1024$.

Method	Keys		No keys	
	inject	error	inject	error
KANERVA	0.000	0.000	0.194	0.000
KANERVA, lexical	0.000	0.000	0.206	0.000
First version	0.000	0.000	0.213	0.026
Hybrid, live only	0.000	0.000	0.997	0.863
Hybrid, newest of twins	0.217	0.000	0.227	0.010
Hybrid RRF	0.997	0.865	0.997	0.863
Hybrid + reranker	0.999	0.702	0.999	0.702
BM25	0.941	0.853	0.941	0.853

Table 8: Prequential recall at $B = 512$, mean ± s.d. over five seeds. "Noisy" flips 20% of rewards.

Dataset	Static	Online	UCB	Noisy	Debiased
LongMemEval-S	0.928 ± 0.000	0.928 ± 0.000	0.928 ± 0.000	0.928 ± 0.000	0.928 ± 0.000
LoCoMo	0.746 ± 0.001	0.725 ± 0.003	0.731 ± 0.005	0.729 ± 0.007	0.728 ± 0.007
RepoMem	0.899 ± 0.001	0.896 ± 0.002	0.896 ± 0.002	0.896 ± 0.001	0.896 ± 0.004

Table 9: Reversible consolidation (LongMemEval-S: 60 random questions). τ is the similarity threshold, R the extractive rate.

Dataset, τ, rate	Clusters	Sources	Tokens saved	Recall@1024
LoCoMo, baseline	0	0	0.0%	0.824
LoCoMo, $\tau = 0.7$, $R = 0.5$	577	1972	23.0%	0.669
LoCoMo, $\tau = 0.8$, $R = 0.5$	195	535	5.6%	0.791
LoCoMo, $\tau = 0.9$, $R = 0.5$	29	71	0.5%	0.823
LongMemEval-S, baseline	0	0	0.0%	0.969
LongMemEval-S, $\tau = 0.7$, $R = 0.5$	7985	20806	39.4%	0.689
LongMemEval-S, $\tau = 0.8$, $R = 0.5$	4625	10327	19.9%	0.789
LongMemEval-S, $\tau = 0.9$, $R = 0.5$	458	1015	2.1%	0.969

Table 10: Forgetting on RepoMem streams: final active store and recall.

Policy	Active items	Active tokens	Recall@1024
keep-all	965	21492	0.923
idle-730d	474	11661	0.857
idle-365d	243	6176	0.713
idle-180d	130	3227	0.551
cap-8k	426	7986	0.750
cap-4k	239	3989	0.540

7 Limitations

No LLM in the evaluation. Every metric is retrieval-level. We do not know how much recall per token translates into correct answers or finished tasks. Evidence recall treats a turn as all-or-nothing, and a selected turn may still be hard for a model to use. Answer accuracy with an LLM is the obvious next measurement.

Baselines that need an LLM were not run. Mem0 with inference, Graphiti, A-MEM, HippoRAG, LangMem and Cognee all call an LLM on write, so the comparison does not cover them. Mem0 ran only in its verbatim mode (`infer=False`), on a subset. Our comparison says nothing about what LLM extraction would add to those systems, and published evidence suggests a small effect on the write side [69].

Generalisation. The relevance model was fitted on four benchmark families, with BEAM-1M as the only source of long conversations, so the held-out result on BEAM is a test on new conversations from a family it has seen. On RepoMem, also a family it has seen, the gain was much smaller. A deployment on notes unlike any of these may behave differently, and the features assume turn structure that real stores may lack. LongMemEval-S and LoCoMo informed the design of the method and are not clean; only the held-out numbers are.

Validity evidence is narrow. The near-twin rule works on RepoMem because versions differ only in a value, and it does not fire on natural-language updates. It also treats look-alike events as versions, which costs recall on code history at large budgets. The hazard model has no ablation that isolates it, and detection of historical intent was not evaluated on any benchmark.

Feedback was simulated. The online experiments derive rewards from evidence labels. Real implicit signals, such as an item's identifiers reappearing in tool calls, are noisier and biased in ways that label flipping does not capture. The plugin's signal has not been validated against real sessions.

RepoMem is a proxy. Queries come from templates over extracted facts, so they contain identifiers and favour lexical search; real prompts are vaguer. The agent's notes are simulated from code and commit history. Every changed fact is assumed to be observed, so a store never misses a change. The task track uses file overlap as relevance, which is a weak label. Keyed mode gets perfect slot keys from the static analyzer, so it is an upper bound on what a real write path provides; plain mode is the conservative end. All repositories are Python libraries, and five repositories give wide intervals.

Answer-term recall. It counts word stems, so it misses paraphrase, and it can credit a word found in an unrelated item. It serves here to compare excerpts with full items, not as a measure of answerability.

Theory versus code. Theorem 2 assumes a linear μ, conditionally sub-Gaussian rewards and validity revealed with feedback. In practice validity is seen only when a successor appears, so unkeyed items that are stale enter learning with a reward of zero. As stated in Section 5, the bound is about scaling, and the defaults differ from the analyzed algorithm. Hazards assume exponential lifetimes within a class. The certificate bounds the surrogate F, not recall, and Proposition 7 bounds expected hits under the model's probabilities, which are calibrated only on data like the training data. Theorem 1 covers selection before the plugins' threshold, not after it. Excerpts have no guarantee.

Plugins. The benchmarks ran the library with exact token counts. The plugins estimate tokens with a dependency-free approximation, and their probability threshold for injection was not evaluated. Their tokenizer also handles non-Latin scripts, which came after the test; on a sample of held-out queries this changed fewer than one row in 300, by less than a thousandth of recall on average. Retrieval quality on languages other than English was not measured.

Token counts. We count with `cl100k_base` as a proxy for Claude's tokenizer, and the 14-token item overhead is an approximation of the rendered prefix.

8 Conclusion

Treating agent memory as budgeted selection over items with explicit validity gives a read path that needs no LLM, and a theory in which expiring items have a clear price. The strongest practical lever turned out to be simpler than the theory: a calibrated estimate of what each candidate is worth. On held-out long conversations this recovers far more of the evidence per token than hybrid retrieval, with or without a cross-encoder. On code history the gain over hybrid retrieval is confined to small budgets and reverses at large ones, where the near-twin rules cost recall. Stale injection is a large problem on code facts, and simple rules fix most of it, so the probabilistic validity model has not yet shown value beyond them. Free-text updates remain unsolved, and online learning adds nothing on top of the fitted model.

The next step is an end-to-end study: whether higher recall per token, and fewer stale injections, change what

a coding agent does. The retrieval results here are necessary for that outcome and do not establish it.

Reproducibility. `make data && make reproduce` rebuilds the results in `results/*.json`, and `make check` verifies each number in this paper against those files. The RepoMem manifest pins every repository snapshot.

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A Proofs

Proof of Lemma 1. For a facet f, $g_f(S) = 1 - \prod_{m \in S}(1 - u_m)$ with $u_m = w(m)a_{mf} \in [0, 1]$. Adding m to S gains $u_m \prod_{n \in S}(1-u_n)$, which is non-negative and non-increasing in S. So g_f is monotone submodular. The additive term is modular with non-negative coefficients. Each term $S \mapsto \max_{m \in S} s(c, m)$ (with $\max \emptyset = 0$) is the facility-location function, which is monotone submodular. Non-negative combinations preserve both properties, and $F(\emptyset) = 0$.

Coordinate monotonicity follows because $\prod(1 - w(m)a_{mf})$ decreases in each $w(m)$ and the third term does not depend on w. For the Lipschitz bound, $|\prod_i(1 - u_i) - \prod_i(1 - v_i)| \le \sum_i |u_i - v_i|$ for $u, v \in [0, 1]$ by telescoping. So each g_f moves by at most $\sum_{m \in S} |w(m) - w'(m)|$ as $a_{mf} \le 1$, and $\sum_f \omega_f = 1$. The additive term moves by at most η times the same sum, and the factor $(1 - \gamma)$ gives L. □

Proof of Proposition 1. Take additive U. One item a has score $1 + \epsilon$ and cost B. There are $\lfloor B/c_{\min} \rfloor$ items of score 1 and cost $c_{\min}$. The rule fills with a first and then has no budget, so it earns $1 + \epsilon$ against OPT $\ge \lfloor B/c_{\min} \rfloor$. Let $\epsilon \to 0$. □

Proof of Proposition 2. Let there be k facets of equal weight. Facet 1 has k duplicates, each with modular score $1 + \epsilon$. Each other facet has one item with score 1. All costs equal 1, and $B = k$. A modular rule takes the k duplicates and covers one facet; the optimum takes one item per facet and covers all k. □

Proof of Proposition 3. Walking from newest to oldest, keep each item that fits. Since the later items cost more than B in total, one of them fails to fit, and at that moment the remaining budget is below its cost, which is at most $c(m)$. The remaining budget only shrinks afterwards, so m never fits. A relevance rule does not depend on the items' order in time. □

Proof of Proposition 4. Take two versions with equal text similarity to the query, and let the assignment of the two timestamps be uniform. An invariant score cannot depend on the assignment, so the ranking between the two is independent of which one is current. The outdated version ranks first with probability 1/2, or at least 1/2 under ties broken independently of time. □

Proof of Proposition 5. Let each round have two items with features e_1, e_2, costs 1, $B = 1$, $\rho \equiv 1$ and additive F. For a fixed θ, assume without loss of generality that $\theta_1 \ge \theta_2$; the fixed policy injects e_1 on every round. Take $\theta^\star = (0, 1)$, which is valid since $\mu \in [0, 1]$. The policy earns 0 per round against 1, so its α-regret is αT for each $\alpha > 0$. Theorem 2 with $\varepsilon \equiv 0$ gives $\tilde{O}(\sqrt{T})$ on this instance. □

Proof of Theorem 1. Let O be an optimal set and consider the density run ($\varrho = 1$) as a sequence of considered items. Let $g_1, g_2, \dots$ be the items accepted, in order, and write $G_i = \{g_1, \dots, g_i\}$. If no element of O is rejected, then every element of O is either accepted or left with zero marginal gain at termination, so $F(G) \ge F(G \cup O) \ge F(O)$ and we are done. Otherwise let $e^\star$ be the first element of O rejected, and let ℓ be the number of accepted items before it, so $c(G_\ell) + c(e^\star) > B$.

For $i \le \ell$, when g_i was chosen, every element of $O \setminus G_{i-1}$ was still available, since none had been rejected. By monotonicity and submodularity,

$$F(O) \le F(O \cup G_{i-1}) \le F(G_{i-1}) + \sum_{o \in O \setminus G_{i-1}} \Delta(o \mid G_{i-1}) \le F(G_{i-1}) + B\,\frac{\Delta(g_i \mid G_{i-1})}{c(g_i)}.$$

The same inequality holds with $e^\star$ in the role of $g_{\ell+1}$ and $G_{\ell+1} := G_\ell \cup \{e^\star\}$, $c(g_{\ell+1}) := c(e^\star)$, since $e^\star$ had maximum density when chosen. Set $\delta_i = F(O) - F(G_i)$. The inequality reads $\delta_i \le (1 - c(g_i)/B)\,\delta_{i-1}$. If some $\delta_i \le 0$ with $i \le \ell$ then $F(G) \ge F(O)$ and we are done. Otherwise, using $0 \le 1 - x \le e^{-x}$ on $x = c(g_i)/B \in (0, 1]$,

$$\delta_{\ell+1} \le F(O) \prod_{i=1}^{\ell+1} \Big(1 - \frac{c(g_i)}{B}\Big) \le F(O) \exp\Big(-\frac{\sum_{i \le \ell+1} c(g_i)}{B}\Big) \le \frac{F(O)}{e},$$

because $\sum_{i \le \ell+1} c(g_i) = c(G_\ell) + c(e^\star) > B$. So $F(G_\ell \cup \{e^\star\}) \ge (1 - 1/e)F(O)$. By subadditivity $F(G_\ell) + F(\{e^\star\}) \ge (1 - 1/e)F(O)$, so one of the two terms is at least $\frac{1}{2}(1 - 1/e)F(O)$. Now $F(G) \ge F(G_\ell)$ by monotonicity, and $F(\{e^\star\})$ is at most the value of the best singleton, which fits since $c(e^\star) \le B$. The algorithm returns the maximum over a set of runs that includes both. □

Proof of Proposition 6. For any feasible O, monotonicity and submodularity give $F(O) \le F(S) + \sum_{m \in O \setminus S} \Delta(m \mid S)$. The indicator of $O \setminus S$ is feasible for the linear program, whose optimum is reached by the fractional-knapsack rule. □

Lemma 2 (Batched elliptical potential)**.** *Let $\lambda \ge K$, $V_1 = \lambda I$, $V_{t+1} = V_t + \sum_{i \in A_t} x_i x_i^\top$ with $|A_t| \le K$ and $\|x_i\| \le 1$. Then $\sum_{t \le T} \sum_{i \in A_t} \|x_i\|^2_{V_t^{-1}} \le 2d \log(1 + KT/(d\lambda))$.*

Proof. Let $B_t = V_t^{-1/2} (\sum_{i \in A_t} x_i x_i^\top) V_t^{-1/2}$, positive semidefinite with $\operatorname{tr} B_t = \sum_{i \in A_t} \|x_i\|^2_{V_t^{-1}} \le K/\lambda \le 1$. So its eigenvalues b_j lie in $[0,1]$, and $\log\det(I + B_t) = \sum_j \log(1 + b_j) \ge \sum_j b_j \log 2 \ge \frac{1}{2} \operatorname{tr} B_t$. Since $\det V_{t+1} = \det V_t \det(I + B_t)$, summing gives $\sum_t \operatorname{tr} B_t \le 2 \log(\det V_{T+1} / \det V_1)$. Finally, $\operatorname{tr} V_{T+1} \le d\lambda + KT$ and AM–GM give $\det V_{T+1} \le (\lambda + KT/d)^d$. □

Proof of Theorem 2. Good event. Order items within a round arbitrarily, and take the filtration that reveals a round's Z values before its rewards. Then $X_{t,m} = Z_{t,m} x_{t,m}$ is predictable, $Z_{t,m} \xi_{t,m}$ is conditionally R-sub-Gaussian, and items with $Z = 0$ contribute nothing. The self-normalized bound [1, Theorem 2], applied to $\theta^\star - \theta_0$ with targets $y - x^\top \theta_0$, gives with probability at least $1 - \delta/4$ for all t: $\|\hat\theta_t - \theta^\star\|_{V_t} \le \beta_T$. The sequence has at most KT terms. So $|x^\top \hat\theta_t - x^\top \theta^\star| \le \beta_T \|x\|_{V_t^{-1}}$ for every x by Cauchy–Schwarz. Combined with the validity event (probability at least $1 - \delta/2$), and writing $\bar\rho_{t,m} = \min(1, \hat\rho_{t,m} + \varepsilon_{t,m})$ and $\bar\mu_{t,m}$ for the clipped upper confidence bound,

$$\rho_{t,m} \le \bar\rho_{t,m} \le \rho_{t,m} + 2\varepsilon_{t,m}, \qquad \mu_{t,m} \le \bar\mu_{t,m} \le \mu_{t,m} + b_{t,m}, \quad b_{t,m} := \min(1, 2\beta_T \|x_{t,m}\|_{V_t^{-1}}).$$

Clipping preserves the lower bound as $\mu_{t,m} \in [0,1]$. In particular $w_t^\star \le \bar w_t$ coordinate-wise.

Per-round regret. Let $S_t^\star$ be optimal for $w_t^\star$. By coordinate monotonicity, the oracle property, and the Lipschitz property,

$$\alpha F_t(S_t^\star; w_t^\star) \le \alpha F_t(S_t^\star; \bar w_t) \le F_t(S_t; \bar w_t) \le F_t(S_t; w_t^\star) + L \sum_{m \in S_t} (\bar w_{t,m} - w_{t,m}^\star).$$

Moreover, $\bar w - w^\star = \bar\rho\bar\mu - \rho\mu = \rho(\bar\mu - \mu) + (\bar\rho - \rho)\bar\mu \le \rho\, b + 2\varepsilon$.

Summation. $b_{t,m}$ is determined by the history before round t, and $\mathbb{E}[Z_{t,m} \mid \text{history}] = \rho_{t,m}$, while Z values within a round may be dependent. So $D_t = \sum_{m \in S_t} (\rho_{t,m} - Z_{t,m}) b_{t,m}$ is a martingale difference sequence with $|D_t| \le K$, and Azuma–Hoeffding gives, with probability at least $1 - \delta/4$, $\sum_t \sum_{m \in S_t} \rho_{t,m} b_{t,m} \le \sum_t \sum_{m \in S_t} Z_{t,m} b_{t,m} + K\sqrt{2T \log(4/\delta)}$. The items with $Z = 1$ are exactly the ones entering V. By Cauchy–Schwarz and Lemma 2,

$$\sum_t \sum_{m \in S_t} Z_{t,m} b_{t,m} \le 2\beta_T \sqrt{KT \sum_t \sum_{m : Z = 1} \|x_{t,m}\|^2_{V_t^{-1}}} \le 2\beta_T \sqrt{2dKT \log(1 + KT/(d\lambda))}.$$

A union bound over the three events finishes the proof, and $\sum_t \sum_{m \in S_t} 2\varepsilon_{t,m}$ is the remaining term. □

Proof of Proposition 8. Write $\lambda = \lambda_k$. The change count $N(s)$ has intensity $\lambda Y(s)$, where Y is the predictable number of items at risk, and $E(s) = \int_0^s Y$. For any fixed real θ, $M_\theta(s) = \exp(\theta N(s) - \lambda(e^\theta - 1) E(s))$ is a non-negative local martingale with $M_\theta(0) = 1$, hence a supermartingale, so $\Pr[M_\theta \ge 2/\delta] \le \delta/2$ at the stopping time by Markov's inequality and optional stopping.

Let $\ell = \log(2/\delta)$ and $\theta = \sqrt{\ell/(\lambda E_0)} = \eta/2 \le \frac{1}{4}$. Taking $+\theta$ and using $(e^\theta - 1)/\theta \le 1 + \theta$ on $[0,1]$: $N \le \lambda E(1 + \theta) + \ell/\theta$. Taking $-\theta$ and using $(1 - e^{-\theta})/\theta \ge 1 - \theta$: $N \ge \lambda E(1 - \theta) - \ell/\theta$. Dividing by λE and using $E \ge E_0$, with $\ell/(\theta \lambda E_0) = \theta$, gives $|\hat\lambda/\lambda - 1| \le 2\theta = \eta$.

Put $s = \hat\lambda/\lambda \in [\frac{1}{2}, \frac{3}{2}]$. By the mean value theorem, for some σ between 1 and s, so $\sigma \ge \frac{1}{2}$, we have $|e^{-s\lambda a} - e^{-\lambda a}| = |s - 1|\, \lambda a\, e^{-\sigma \lambda a} \le \eta/(e\sigma) \le 2\eta/e$, using $\sup_{u \ge 0} u e^{-\sigma u} = 1/(e\sigma)$, for all a at once. □

Proof of Proposition 9. Let each round have m items with equal features, $\mu = 1$, $c = 1$, $B = 1$ and $F(S; w) = \sum_{i \in S} w_i$, with the index of the valid item uniform and independent across rounds. The information a blind learner gets from past rounds is independent of the current valid index, so whichever item it picks is valid with probability $1/m$ and $\mathbb{E} F_t(S_t) = 1/m$, while $\mathrm{OPT}_t = 1$. □

Proof of Proposition 10. $\int_0^\infty e^{-\lambda a} \frac{\beta^\alpha \lambda^{\alpha-1} e^{-\beta\lambda}}{\Gamma(\alpha)}\, d\lambda = (\beta/(\beta + a))^\alpha$. The logarithm is $-\alpha \log(1 + a/\beta) = -\alpha \log a + \alpha \log \beta + o(1)$. A single access at age a has ACT-R activation $B = \log a^{-d} = -d \log a$. □

Proof of Theorem 3. Φ is monotone submodular (facility location), $\Phi(\emptyset) = 0$ and $\Phi(X) = 1$. For the greedy sequence with density-maximal g_i, as in the proof of Theorem 1, $\Phi(Z^\star) - \Phi(Z_{i-1}) \le R^\star \Delta_i / c(g_i)$, so $d_i := \Phi(Z^\star) - \Phi(Z_i) \le (1 - c(g_i)/R^\star) d_{i-1} \le e^{-c(g_i)/R^\star} d_{i-1}$ whenever $d_{i-1} > 0$, using $1 - x \le e^{-x}$ for all real x. If $d_{i-1} \le 0$ for some i, then $\Phi(Z_\ell) \ge \Phi(Z^\star)$ by monotonicity. In every case $\Phi(Z_\ell) \ge (1 - e^{-R_\ell/R^\star})\Phi(Z^\star)$. Substituting $\Phi = 1 - D$ gives the first claim.

For the second, unit vectors satisfy $\|e_x - e_z\|^2 = 2(1 - \langle e_x, e_z \rangle)$. So $|\langle e_q, e_x - e_z \rangle| \le \sqrt{2(1 - s(x, z))}$, which bounds the score change by $\ell_{\mathrm{u}} \sqrt{2(1 - s(x, z(x)))}$. Average over p and apply Jensen's inequality to the square root, noting that $\mathbb{E}_p[1 - s(x, z(x))] = D(Z_\ell)$. □

Proof of Theorem 4. Condition on r and let T be the time until the first gap of length τ in a Poisson process of rate r started at the arrival. Let $A \sim \mathrm{Exp}(r)$ be the first access. If $A > \tau$ then $T = \tau$, otherwise $T = A + T'$ with T' an independent copy of T by the memoryless property. So $\mathbb{E}T = \mathbb{E}\min(A, \tau) + (1 - e^{-r\tau})\mathbb{E}T$. With $\mathbb{E}\min(A, \tau) = (1 - e^{-r\tau})/r$, this gives $\mathbb{E}T = (e^{r\tau} - 1)/r$.

The pairs (arrival time, lifetime) form a marked Poisson process with i.i.d. marks. By the marking theorem, the number of items alive at t is Poisson with mean $a \int_0^t \Pr[L > u]\, du$, which converges to $a\,\mathbb{E}L$ under the finiteness assumption. □

Proof of Proposition 7. For any set S, $|\sum_{m \in S} p_m - \sum_{m \in S} \hat{p}_m| \le e(S)$. Hence $\sum_{\hat{S}} p \ge \sum_{\hat{S}} \hat{p} - e(\hat{S}) \ge \alpha \sum_{S^\star} \hat{p} - e(\hat{S}) \ge \alpha \sum_{S^\star} p - \alpha e(S^\star) - e(\hat{S})$, where the middle step uses that $S^\star$ is feasible. For $\alpha = 1$, the items in $S^\star \cap \hat{S}$ cancel: apply the same chain to $S^\star \setminus \hat{S}$ and $\hat{S} \setminus S^\star$, using $\sum_{\hat{S} \setminus S^\star} \hat{p} \ge \sum_{S^\star \setminus \hat{S}} \hat{p}$, which holds because $\hat{S}$ is optimal for $\hat{p}$.

For $\alpha = \frac{1}{2}$ with an additive objective: let density greedy stop at the first item g that does not fit, with accepted set G, ignoring later items, which only add value. The items are in density order, so the value of $G \cup \{g\}$ is at least the optimum of the fractional relaxation, and so at least the integer optimum. Thus $\hat{p}(G) + \hat{p}(g) \ge \mathrm{OPT}$, and the better of G and the best singleton has value at least $\frac{1}{2}\mathrm{OPT}$. □